# Activity 9: Murder Rate Across the United States (1976)

**Course: DSA0601 – Data Handling and Visualization**

**Slot: C**

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## 1.) Base map — Choropleth

Aim: To visualize murder rates per 100,000 people across the contiguous United States in 1976, using the cividis colour scale.

R code (run after the given setup code):

```
library(ggplot2)

library(maps)

ggplot(mapdat, aes(long, lat, group = group, fill = murder)) +

  geom_polygon(colour = "white") +

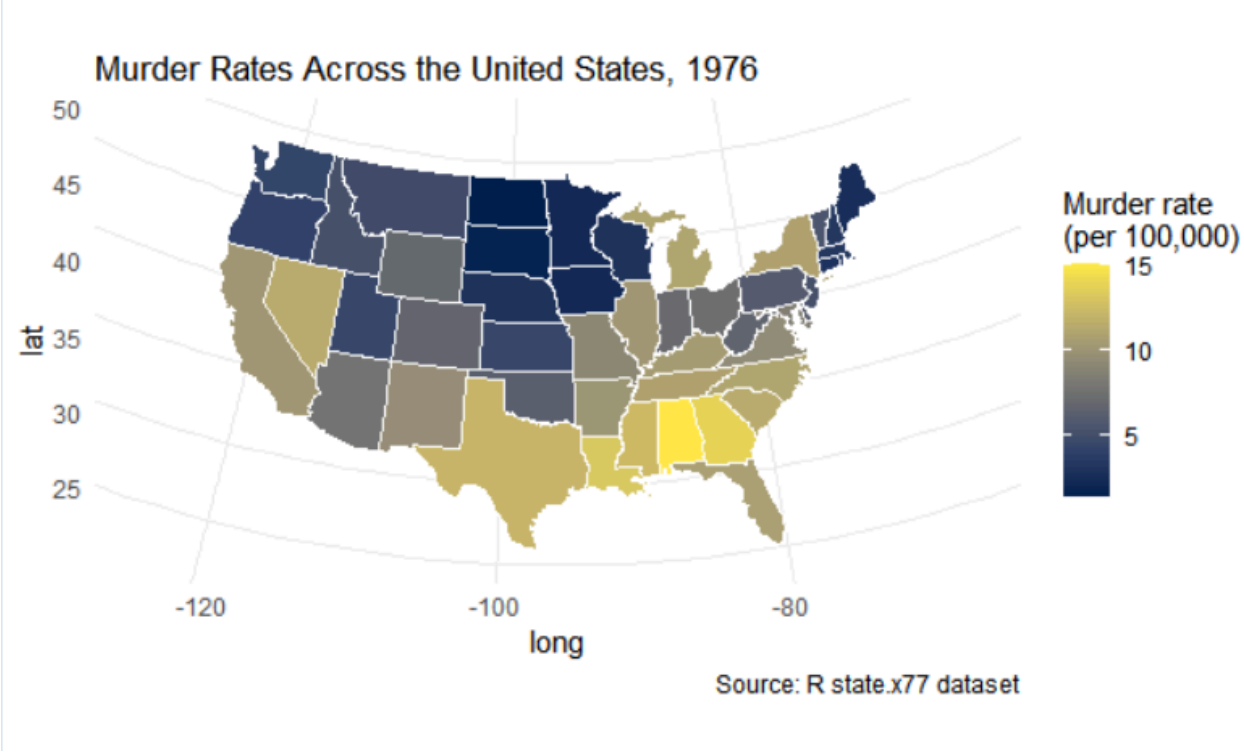
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +

  scale_fill_viridis_c(option = "cividis") +

  labs(
```

```
  title = "Murder Rates Across the United States, 1976",
  fill = "Murder rate\n(per 100,000)",
  caption = "Source: R state.x77 dataset"
 ) +
 theme_minimal()
```

## Output



The map uses colour intensity to show differences in murder rates. A clear legend helps readers compare states.

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## 2. Regional story

Use the following R code to calculate the regional means and correlation:

```
# Mean murder rate for each region

regional_mean <- tapply(V, d$zone, mean)

print(regional_mean)

# Highest and lowest regional means

print(max(regional_mean))

print(min(regional_mean))

print(max(regional_mean) - min(regional_mean))



# Correlation between murder rate and income

print(cor(V, W))
```

| Region | Mean murder rate |
|---|---|
| Northeast | 4.722 |
| South | 10.581 |
| North Central | 5.275 |
| West | 7.215 |

- Highest region: South, with a mean of 10.581.
- Lowest region: Northeast, with a mean of 4.722.
- Gap: $10.581 - 4.722 = 5.859$ murders per 100,000 people.
- Correlation: $r \approx -0.230$.

**Interpretation:** Murder rate and per-capita income have a weak negative correlation because the absolute correlation is below 0.3; this association does not prove that income causes murder rates to increase or decrease.

## Output

```
> # Highest and lowest regional means
> print(max(regional_mean))
[1] 10.58125
> print(min(regional_mean))
[1] 4.722222
> print(max(regional_mean) - min(regional_mean))
[1] 5.859028
>
> # Correlation between murder rate and income
> print(cor(V, W))
[1] -0.2300776
```

## 3. Classes change the story

R code for creating both five-class maps:

```
library(ggplot2)
library(maps)

# Equal-interval classes
equal_class <- cut(V, breaks = 5)

# Quantile classes
quantile_class <- cut(
  V,
  breaks = quantile(V, seq(0, 1, length.out = 6)),
  include.lowest = TRUE
)

# Add classes to the data frame
d$equal_class <- equal_class
d$quantile_class <- quantile_class

# Create map data with equal-interval classes
map_equal <- merge(
  map_data("state"), d,
  by.x = "region", by.y = "id"
)
```

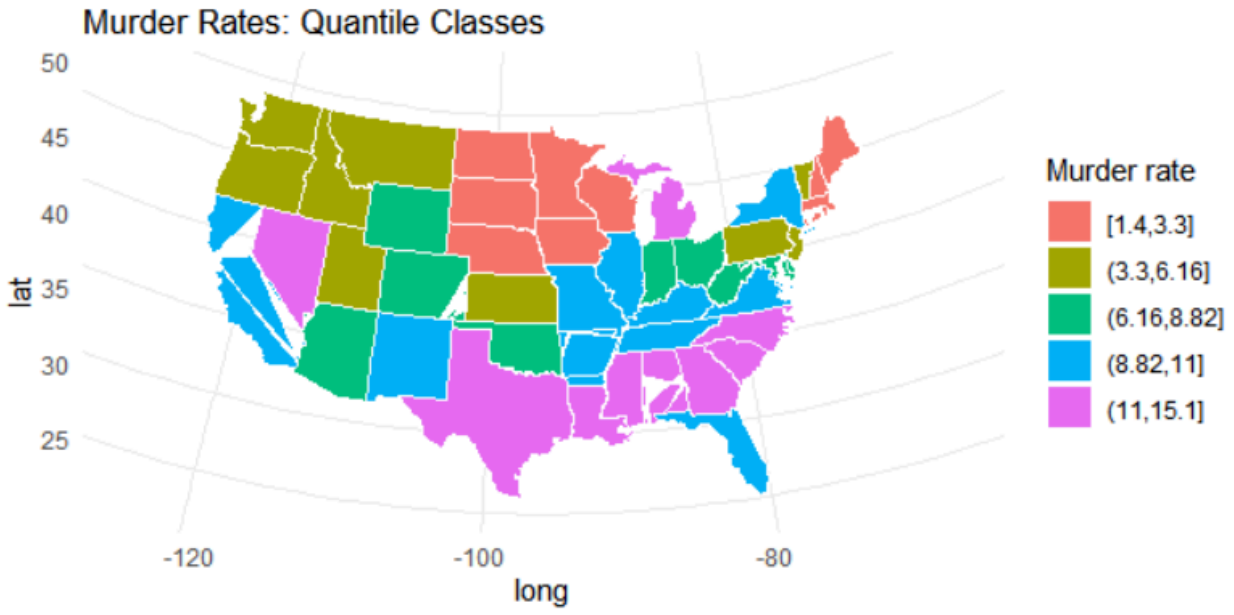
Results using the supplied murder-rate values:

| Method | Lower limit of top class | Number of states in top class |
|---|---|---|
| Equal interval | 12.36 | 4 |
| Quantile | 11.02 | 10 |

- Equal interval makes the highest values look rarer because only four states fall in the highest class.
- Quantile places approximately equal numbers of states in each class, making it easier to compare relative rankings.

My choice: I would use equal intervals for a news website when the aim is to emphasize states with exceptionally high murder rates, while clearly displaying the class boundaries so readers can interpret the map honestly.

## Output



## 4. Counts versus rates

R code:

# Calculate estimated number of murders

count <- d$pop$ * 1000 * $d$murder / 100000

# Top 3 states by murder count

count_top3 <- order(count, decreasing = TRUE)[1:3]

print(data.frame(

State = d$state[count_top3], $Murder_Count = round(count[count_top3]))) #Top3statesbymur$
murder, decreasing = TRUE)[1:3]

print(data.frame(

State = d$state[rate_top3], $Murder_Rate = d$murder[rate_top3]

))

# Number of states common to both lists

length(intersect(d$state[count_top3], $d$state[rate_top3]))

## Results:

**Results:**

| Rank | Top states by estimated count | Top states by murder rate |
|---|---|---|
| 1 | California — 2,183 | Alabama — 15.1 |
| 2 | New York — 1,969 | Georgia — 13.9 |
| 3 | Texas — 1,493 | Louisiana — 13.2 |

- States appearing in both lists: 0.

Explanation: A map of raw counts can mislead because highly populated states may have more murders simply due to their larger populations; a per-capita murder-rate map is more suitable for comparing risk across states.

## Output

```
      State Murder_Count
1 California         2183
2   New York         1970
3      Texas         1493
>
> # Top 3 states by murder rate
> rate_top3 <- order(d$murder, decreasing = TRUE)[1:3]
> print(data.frame(
+   State = d$state[rate_top3],
+   Murder_Rate = d$murder[rate_top3]
+ ))
      State Murder_Rate
1   Alabama        15.1
2   Georgia        13.9
3 Louisiana        13.2
>
> # Number of states common to both lists
> length(intersect(d$state[count_top3], d$state[rate_top3]))
[1] 0
```

## 5. Tell the story for the North Central region

Calculated results

- North Central mean murder rate: 5.275 per 100,000 people.
- National mean murder rate: 7.378 per 100,000 people.
- Ratio to national mean: $5.275/7.378 \approx 0.715$, or approximately 71.5%.
- Highest North Central state: Michigan, with 11.1 murders per 100,000 people.
- North Central states above the national median of 6.85: 5 states — Illinois, Indiana, Michigan, Missouri and Ohio.

## Headline (12 words maximum)

North Central Murder Rate Below National Average, but Michigan Stands Out

**Evidence sentences**

1. The North Central region recorded an average murder rate of 5.275 per 100,000 people, approximately 28.5% below the national mean of 7.378.
2. Michigan had the region's highest rate at 11.1, and five North Central states exceeded the national median of 6.85.

Recommendation for the news website: Publish the five-class cividis choropleth with a readable legend, a clear source note, and a short North Central regional comparison; emphasize rates rather than raw counts to avoid misleading readers.

**Quick R verification code**

Run this block after the supplied setup to verify the key numerical answers:

```
# Overall statistics

max(dmurder)dstate[which.max(dmurder)]min(dmurder)

dstate[dmurder == min(dmurder)]median(dmurder)

mean(dmurder)#Regionalstatisticstapply(V, dzone, mean)

cor(V, W)

# Class statistics

equal_class <- cut(V, breaks = 5)

quantile_breaks <- quantile(V, seq(0, 1, length.out = 6))

quantile_class <- cut(V, quantile_breaks,

                      include.lowest = TRUE)


tail_equal <- max(V[equal_class == levels(equal_class)[5]])

tail_quantile <- max(V[quantile_class == levels(quantile_class)[5]])
```

print(quantile_breaks)

print(table(equal_class))

print(table(quantile_class))

# North Central story

nc <- d $zone == "NorthCentral"mean(V[nc])mean(V[nc])/mean(V)d$ state[nc][which.max(V[nc])]

sum(V[nc] > median(V))

Source note: The analysis uses R's built-in `state.x77` dataset, which records population estimates from 1975, per-capita income from 1974 and murder rates from 1976.

**Output**

```
> print(quantile_breaks)
   0%   20%   40%   60%   80%  100%
 1.40  3.30  6.16  8.82 11.02 15.10
> print(table(equal_class))
equal_class
(1.39,4.14] (4.14,6.88] (6.88,9.62] (9.62,12.4] (12.4,15.1]
         11          14           7          14           4
> print(table(quantile_class))
quantile_class
  [1.4,3.3]   (3.3,6.16] (6.16,8.82]   (8.82,11]   (11,15.1]
         11           9          10          10          10
>
> # North Central story
> nc <- d$zone == "North Central"
> mean(V[nc])
[1] 5.275
> mean(V[nc]) / mean(V)
[1] 0.7149634
> d$state[nc][which.max(V[nc])]
[1] "Michigan"
> sum(V[nc] > median(V))
[1] 5
```

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Code of Conduct: I certify that this work is my own.

Signature: ***Udhaya kumar U***